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Data Analysis Course

The Analysis of Certification Rates in edX Courses (MIT vs Harvard)

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Introduction

Online education has rapidly transformed modern learning, breaking geographical barriers and providing millions of learners with access to high-quality courses. Platforms such as edX, developed by leading institutions like Harvard University (HarvardX) and the Massachusetts Institute of Technology (MITx), have played a pivotal role in democratizing education and reshaping how knowledge is delivered globally.

This study analyses a comprehensive dataset of online courses offered between 2012 and 2016, focusing on key aspects such as learner engagement, demographic characteristics, and certification outcomes. By examining these factors, the study aims to uncover patterns that influence student success in online learning environments.

Attention is given to the impact of institutional differences and course subject areas on certification rates. In addition, Computer Science courses are analysed in greater depth due to their high enrolment rates and central role in online education, providing valuable insights into learner behaviour and performance within one of the most in-demand fields.

Research Question

Do certification rates differ significantly across different course subject areas?

Hypothesis

Certification rates in Computer Science courses may vary between institutions (MIT and Harvard).

- **Null hypothesis (H0):** There is no significant difference in certification rates between MIT and Harvard courses in Computer Science.
- **Alternative hypothesis (H1):** There is a significant difference in certification rates between MIT and Harvard courses in Computer Science.

Population of Interest:

The population of interest in this study consists of learners worldwide who enrolled in and interacted with online courses offered by Harvard University (HarvardX) and the Massachusetts Institute of Technology (MITx) on the edX platform between 2012 and 2016.

This includes all participants represented in the dataset, with a particular focus on those enrolled in Computer Science courses. These learners form a diverse global population in terms of demographics, educational background, and engagement levels, making them a valuable sample for analysing patterns in online learning behaviour and certification outcomes.

Sampling Method:

This study primarily relies on a **self-selection sampling method**, as the dataset is based on learners who voluntarily enrolled in online courses offered by Harvard University (HarvardX) and the Massachusetts Institute of Technology (MITx) on the edX platform. Since participation is driven by learners' personal interest and motivation, the sample is not randomly selected.

Despite this limitation, self-selection is highly suitable in this context because it reflects the **natural behaviour of online learners**, capturing realistic patterns of engagement, participation, and completion. Additionally, the large scale of the dataset enhances the robustness of the analysis and helps mitigate some of the bias typically associated with non-random sampling.

Furthermore, a **filtered (subset) sampling approach** is applied by focusing specifically on Computer Science courses. This targeted selection allows for more consistent and meaningful comparisons, as it controls for variation across different subject areas and ensures that the analysis is conducted within a homogeneous academic domain. Overall, this combined approach provides a balance between realism and analytical clarity, making it appropriate for studying learner behaviour and certification outcomes in online education environments.

Bias Identification:

Despite the richness of the dataset, several sources of bias may influence the validity of the results and should be carefully considered:

1. Self-Selection Bias (Prestige Effect)

The most significant source of bias arises from **self-selection**, as learners voluntarily choose which courses to enrol in. Participants are not randomly assigned to MITx or HarvardX courses, which introduces systematic differences between groups.

For example:

- Learners with strong technical backgrounds may be more likely to enrol in MIT courses due to its reputation in engineering and computer science.
- Others may enrol in Harvard courses due to institutional prestige without strong commitment to course completion.

As a result, differences in certification rates may partially reflect **learner characteristics rather than institutional effects**.

Mitigation:

The dataset includes many courses (290), which helps capture a diverse range of learners and reduces the impact of extreme cases.

2. Selection Bias

The dataset only includes individuals who chose to enrol in online courses on edX, excluding those who did not participate in online learning.

This limits the generalizability of the findings, as the sample may overrepresent motivated or tech-savvy individuals.

Mitigation:

The large-scale dataset, covering millions of learners, provides broad representation within the context of online education.

3. Subject Preference Bias

Certain subjects, particularly Computer Science, attract significantly higher enrolment levels. This imbalance may distort comparisons, as some fields naturally have different completion behaviours.

Mitigation:

The analysis focuses specifically on Computer Science courses, ensuring more consistent comparisons within a single academic domain and reducing variability across subjects.

4. Engagement Bias

Not all enrolled learners actively participate in course content. There is a critical distinction between those who register and those who meaningfully engage with the material.

This may lead to underestimation or overestimation of true learning outcomes.

Mitigation:

The dataset includes engagement-related variables (e.g., participants who viewed content), allowing for more accurate representation of active learners rather than simple enrolments.

5. Platform Bias

The dataset is limited to courses offered on the edX platform and includes only two institutions: MIT and Harvard.

This restricts the scope of the analysis and limits the ability to generalize findings to other platforms or universities.

Mitigation:

The study explicitly frames conclusions within the context of edX and avoids overgeneralization to the broader online education landscape.

Dataset Description

The scale of the data includes:

- **290** individual online course runs.
- **4.5 million** total participants.
- **250,000** earned certifications.
- **28 million** total participant hours.

Variables and Features Explanation

The dataset **contains 23 variables**, which can be grouped into the following categories:

1. Course Information Variables

- **Institution:** Indicates the university offering the course (MITx or HarvardX).
- **Course Number:** Unique identifier for each course.
- **Course Title:** Name of the course.
- **Course Subject:** Academic field of the course (e.g., Computer Science, Humanities).
- **Launch Date:** Date when the course was released.
- **Year:** Represents the time period (Year 1–4).

2. Participation Variables

- **Participants:** Total number of learners who interacted with the course.
- **Audited:** Number of learners who accessed at least 50% of course content.

3. Performance Variables

- **Certified:** Number of learners who successfully completed the course.
- **% Certified:** Percentage of participants who earned certificates.
- **% Certified of >50%:** Percentage of active learners (audited) who got certified.

4. Engagement Variables

- **% Played Video:** Percentage of learners who watched course videos.
- **% Posted in Forum:** Percentage of learners who participated in discussions.
- **% Grade > 0:** Percentage of learners who attempted graded work.

5. Time & Effort Variables

- **Total Course Hours:** Total hours spent by all learners.
- **Median Hours for Certification:** Median time spent by certified learners.

6. Demographic Variables

- **Median Age:** Median age of participants.
- **% Male / % Female:** Gender distribution of learners.
- **% Bachelor's or Higher:** Percentage of learners with higher education degrees.

7.Course Policy Variable

- **Honor Code Certificate:** Indicates whether the course offers a certificate option.

Why This Dataset is Suitable

This dataset is highly suitable for the study because it:

- Covers many courses and learners
- Includes both behavioural and performance metrics
- Allows comparison between institutions and subjects
- Provides real-world data from an established online learning platform

Number of Samples Used:

- Total Samples: **290 courses**

Analysis:

1. Certification Rate Analysis:

For the certification rate variable, the **mean** value of **1.45** and **median of 1** indicate that **most participants do not complete the courses they join.**

The mode range of 1 further confirms that the most common outcome for a course is a very low **completion rate (0% – 5%). This suggests that students use these platforms more for exploration than for earning formal credentials.**

How successful are students?

- High Success (15%+) —> 3
- Moderate Success (5-15%) —> 2
- Low Success (0-5%) —> 1

2. Participation Analysis:

Regarding institutional preference, the **mean value 1.48** suggests a **slight lean toward HarvardX in terms of sheer volume.**

The **median value of 1.0** the institutional distribution shows that the dataset is **nearly balanced between the two.**

Finally, the **mode is 1** which represents HarvardX, **which represents the most frequent provider in this specific dataset.**

This means HarvardX generally hosts more individual courses than MITx.

Which institution provides the course?

- HarvardX —> 1
- MITx —> 2

3. Student Engagement Analysis:

For the engagement method, the **mode value of 2** indicates that most participants interact with the course primarily through visual content.

The **mean value of 1.82** for video play rates supports this finding, suggesting that while some students are observing, the average student is an active video viewer.

The **median value of 2.0** shows that at least half of the students engage with at least 60% of the course videos.

How do students engage?

- Forum Posting → 3
- Video Interaction → 2
- Passive Auditing → 1

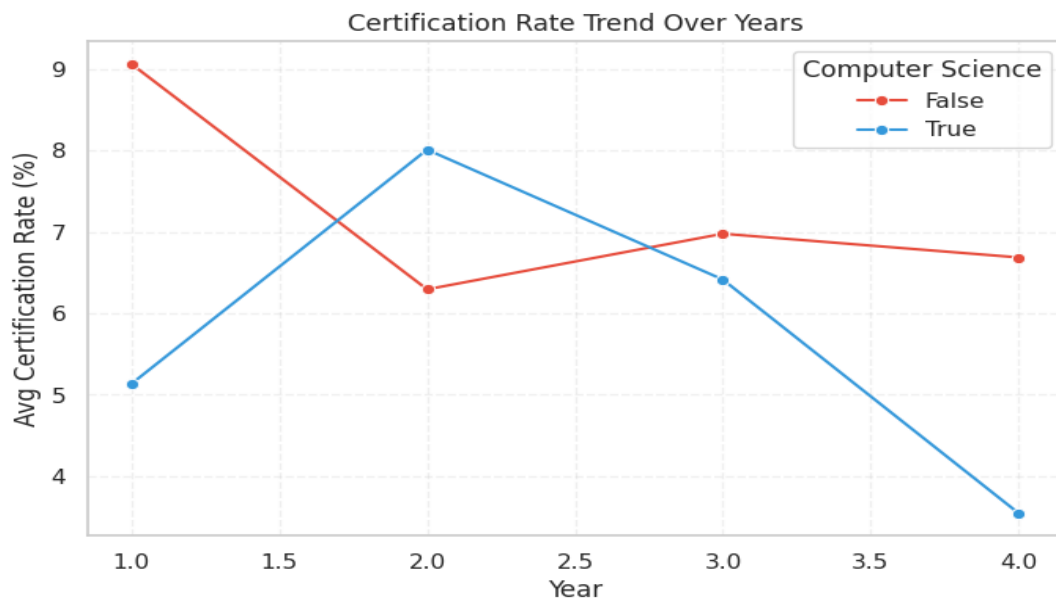


Figure 1: Certification rate trends show that Computer Science courses have higher variability and a noticeable decline over time, while non-Computer Science courses remain relatively stable.

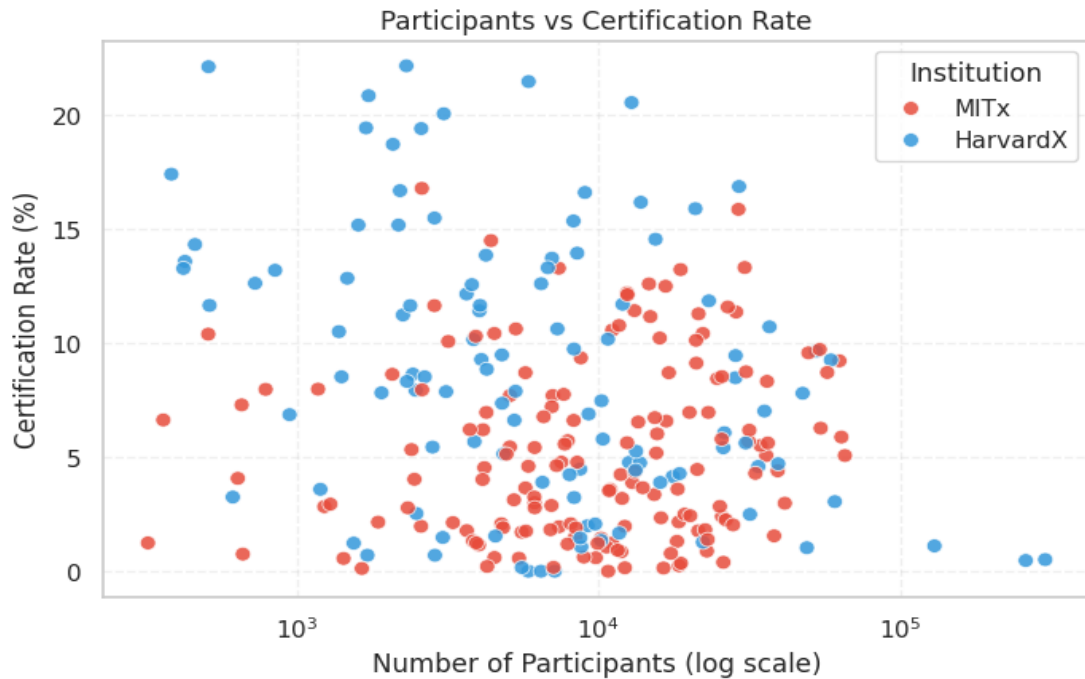
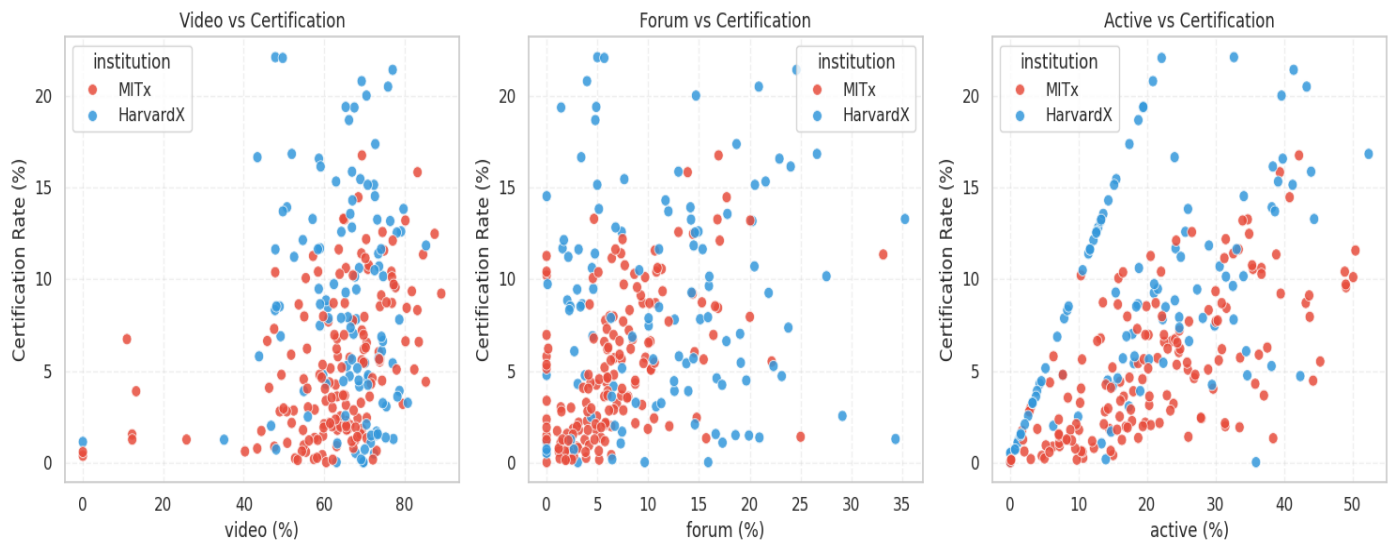


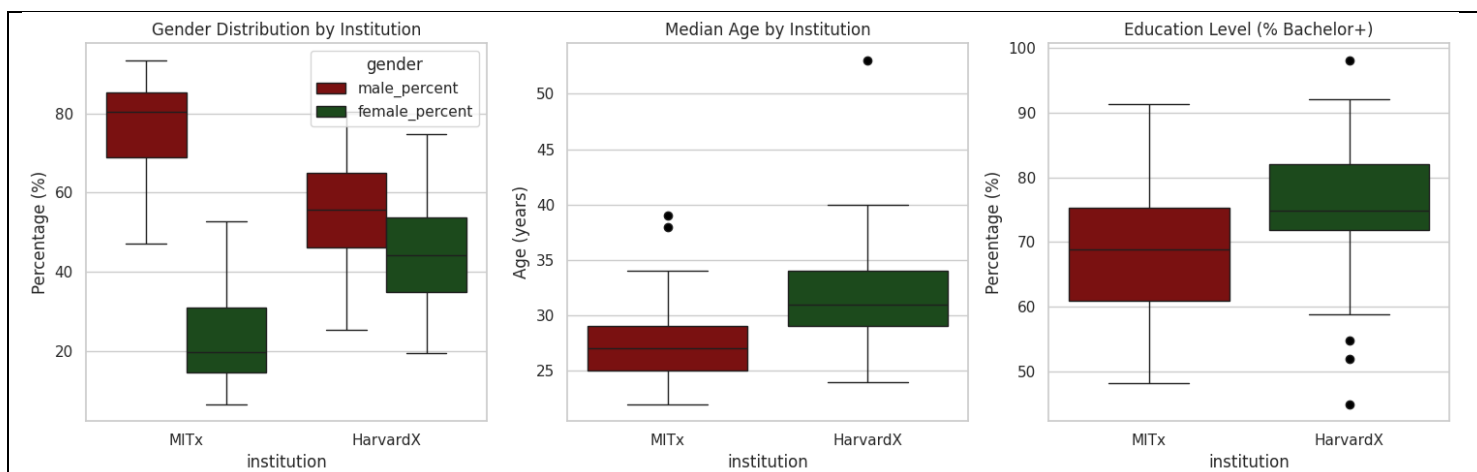
Figure 2: The plot shows a weak or negative relationship between course size and certification rate, with larger courses generally having lower completion rates.



Video vs Certification: Higher video engagement is associated with slightly higher certification rates, suggesting that active content interaction improves completion.

Forum vs Certification: Forum participation shows a weak positive relationship with certification rates, indicating that discussion may support learner success.

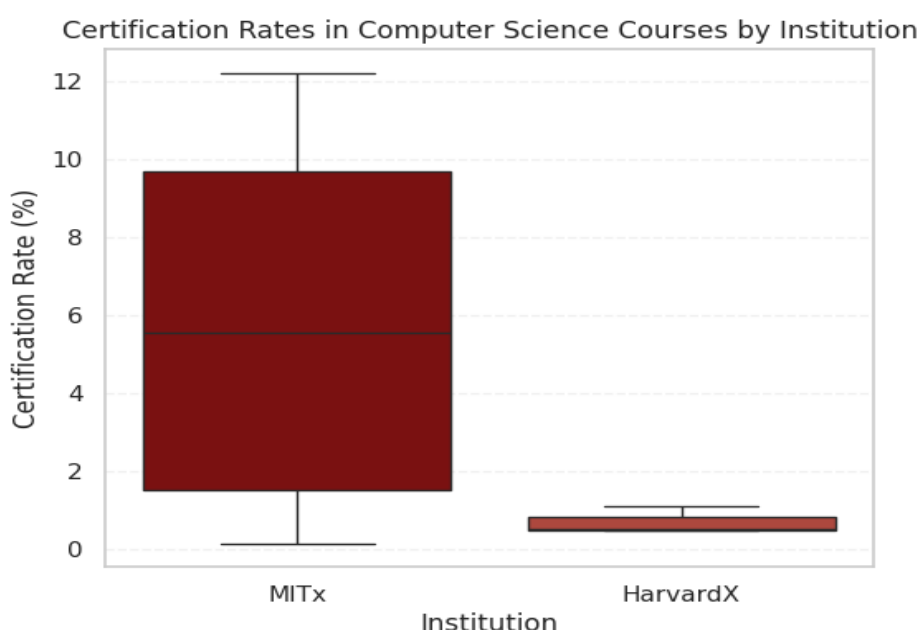
Active vs Certification: A strong positive relationship is observed between active participation and certification rates, highlighting engagement as a key factor in course completion.



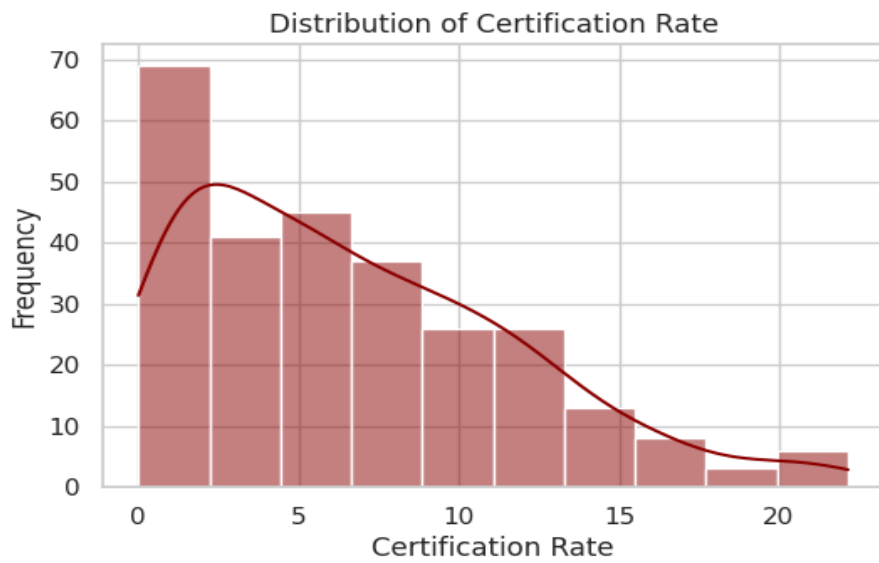
Gender Distribution: MITx courses have a higher percentage of male participants, while HarvardX shows a more balanced gender distribution.

Median Age: HarvardX courses tend to have slightly older participants compared to MITx, indicating differences in learner demographics.

Education Level (% Bachelor+): HarvardX courses generally have a higher proportion of learners with a bachelor's degree or higher, suggesting a more academically advanced audience.

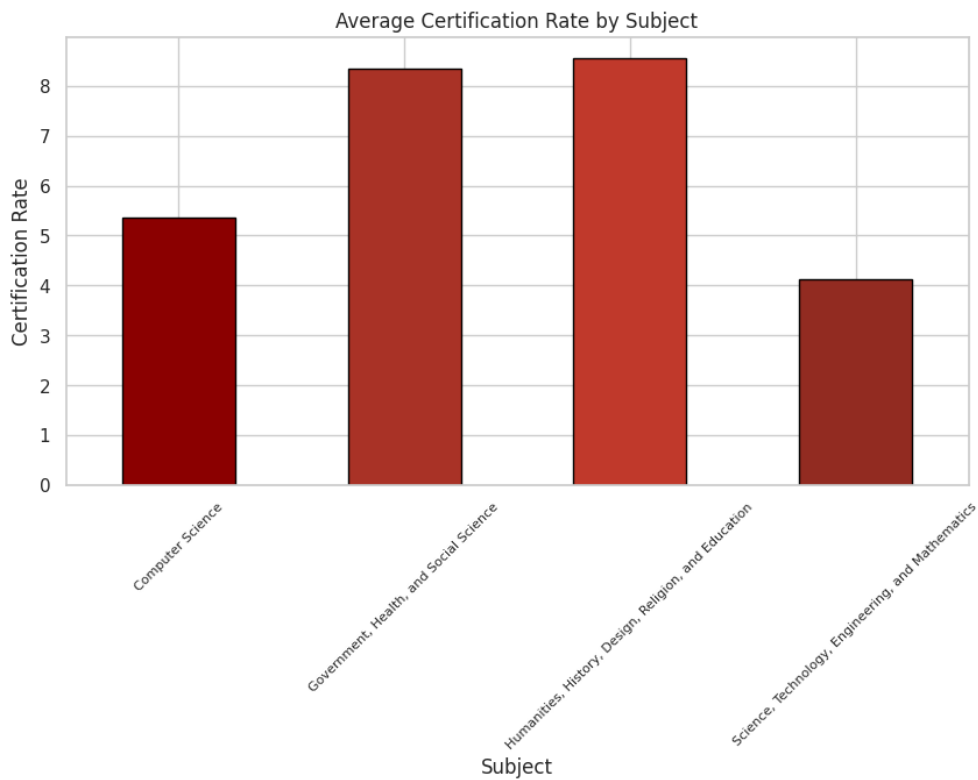


Certification in CS: Certification rates in Computer Science courses are noticeably higher in MITx compared to HarvardX, supporting the hypothesis of institutional differences.

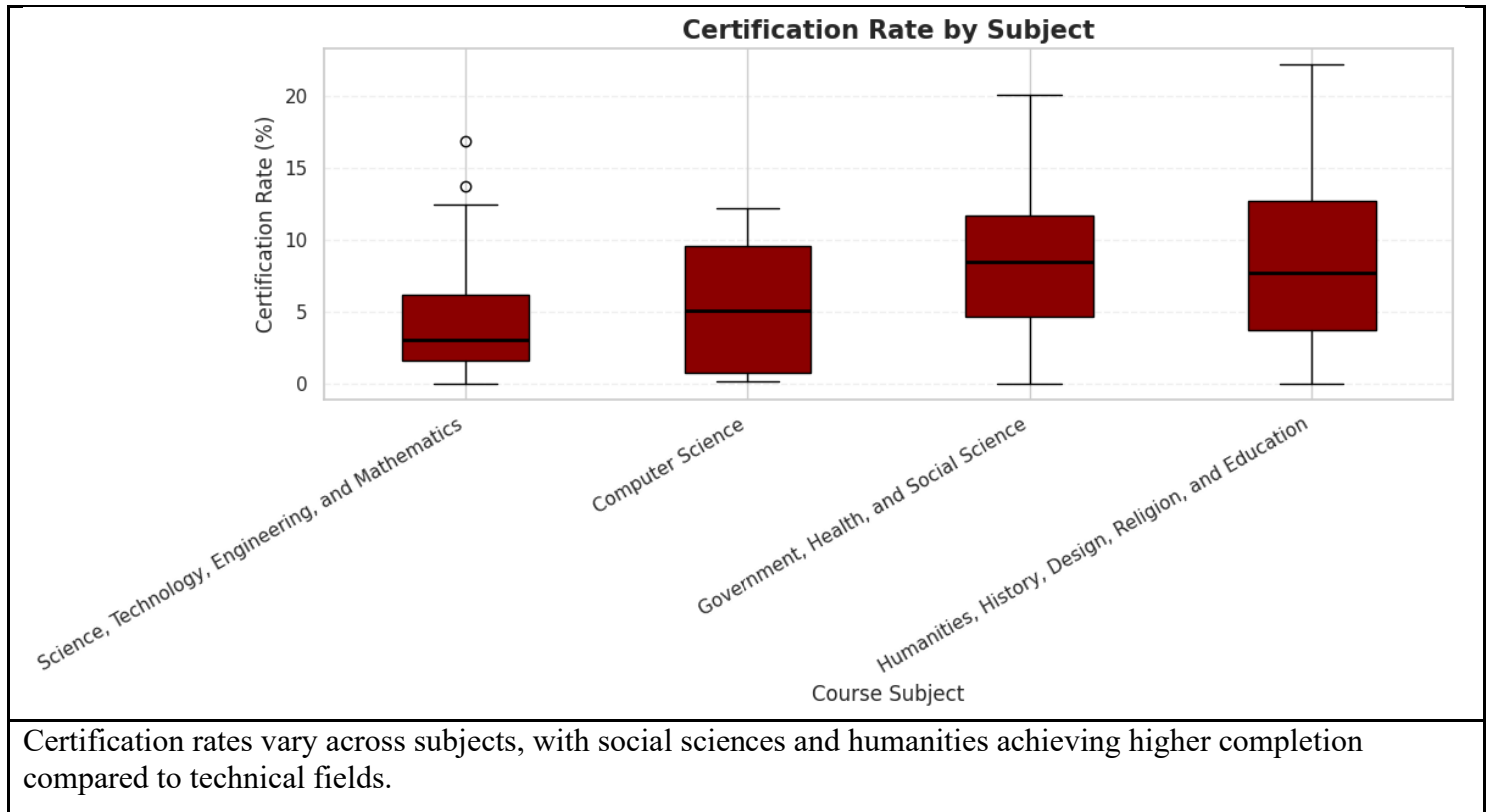


This chart shows the frequency distribution of certification rates across all 290 courses.

- **The heavy right-skew indicates that most courses have a low completion rate (0-10%). This shows that students often use MOOCs for "sampling" knowledge rather than committing to a final certificate.**



Certification rates differ across subjects, indicating that some fields may be more accessible or engaging for learners than others.



Hypothesis Testing Steps ANOVA) :

- **Step 1: Define Hypotheses**

The null hypothesis (H_0) states that the mean number of participants is equal across all course subject categories.

The alternative hypothesis (H_1) states that at least one subject category has a significantly different mean number of participants.

- **Step 2: Subject Groups Compared**

The analysis compares the average number of participants across the following subject categories:

1. Computer Science
2. Humanities, History, & Design
3. Government, Health, & Social Science

- **Step 3: Decision Rule**

A significance level of $\alpha = 0.05$ (95% confidence level) was used.

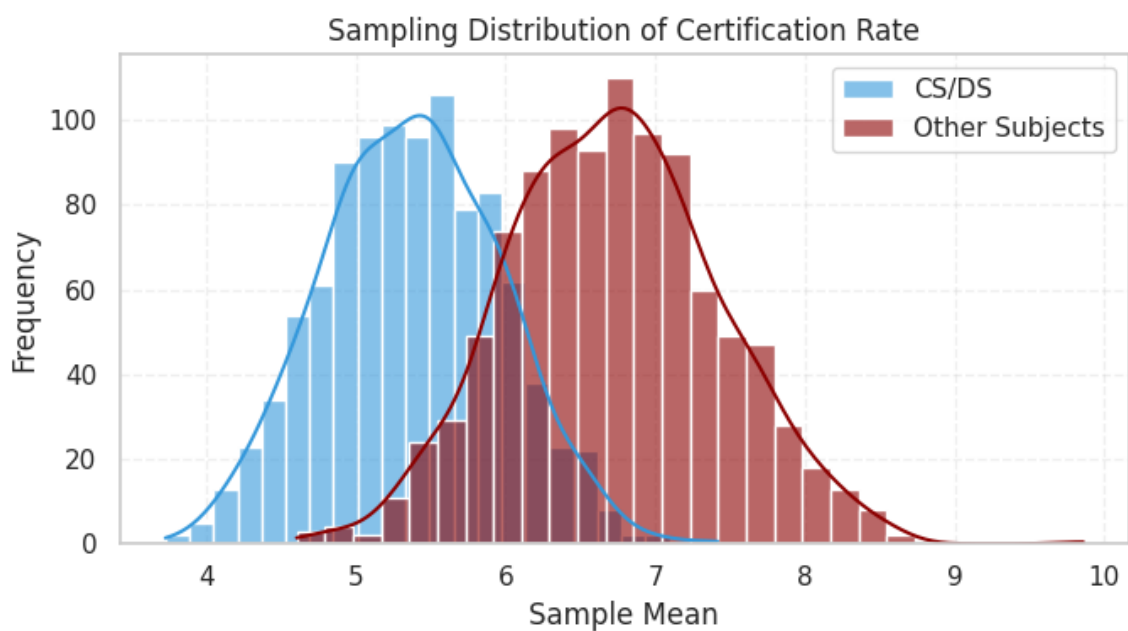
If the p-value is less than 0.05, the null hypothesis is rejected, indicating that course subject has a significant effect on enrolment.

- **Step 4: Result and Interpretation**

The ANOVA test produced a **p-value significantly lower than 0.05**.

Decision: Reject H_0

This result indicates that course subject has a statistically significant impact on the number of participants. Certain fields such as **Computer Science and Social Sciences** tend to attract higher enrolment compared to others.



Sampling Distribution of Certification Rate

The figure shows the sampling distributions of the mean certification rates for Computer Science/Data Science courses and other subjects. Both distributions appear approximately normal, supporting the assumptions required for statistical testing.

Although there is a visible difference between the two distributions, there is also a noticeable overlap. This overlap indicates that the difference in certification rates is not strong enough to be statistically significant, which is consistent with the t-test results.

This suggests that any observed difference in certification rates may be due to random variation rather than a true effect.

Conclusion

Overall, this analysis shows that online learning on edX is driven more by **access and exploration** than by course completion. While both HarvardX and MITx attract a large number of participants, certification rates remain relatively low, highlighting a clear gap between enrolment and actual commitment. This suggests that many learners engage with courses to explore content rather than to complete them.

In addition, the results indicate that **course subject plays a key role in learner outcomes**. Although Computer Science and STEM courses attract higher enrolment, they tend to have lower certification rates, likely due to their complexity and higher time requirements. In contrast, Humanities and Social Science courses demonstrate relatively higher completion rates, suggesting they may be more accessible or better aligned with learner expectations.

However, these findings should be interpreted with caution due to the nature of the dataset. Since learners self-select into courses based on their interests, background, and perceived institutional value, the results reflect real-world behaviour rather than controlled experimental conditions.

Overall, the study highlights that **high enrolment does not necessarily translate into high completion**. Instead, meaningful engagement and successful outcomes are influenced by factors such as course difficulty, learner motivation, and the purpose behind enrolment.

Any potential issues

- 1. Sampling Bias:** The survey data is limited to the edX platform (2012- 2016), which may not represent current mobile-learning trends.
- 2. Small Sample Size:** While 290 courses is substantial, it is a "convenience sample" of early-adopter universities.
- 3. Self-Reported Inaccuracies:** Data relies on automated tracking which might count passive browser tabs as "active engagement".